Case H — Ms. Henderson

19F · Competitive Gymnast · 8-month sharp LBP at L5, extension-provoked

Musculoskeletal Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Painful and restricted lumbar extension
- Paraspinal muscle guarding/spasm secondary to pain
- Unable to perform gymnastics skills requiring extension (walkovers, back handsprings, beam skills)
- Reduced training capacity and skill progression

■■ RED FLAGS

- Nutritional habits and energy:
- "How is your nutrition — do you feel you have enough fuel for training?"
- Radicular symptoms:
- "Have you had any pain, numbness, or tingling travelling down your legs? Any weakness?"
- ■ RED-S / Female Athlete Triad
- Relative Energy Deficiency in Sport
- Significance:
- Screen with LEAF-Q questionnaire; refer to sports dietitian and sports physician/endocrinologist; DEXA scan for BMD; notify team physician. Do not dismiss as simply a training injury.

■ PATHOPHYSIOLOGY (KEY POINTS)

- Spondylolysis is a stress fracture of the pars interarticularis — the isthmus of bone connecting the superior and inferior articular processes of a vertebra.
- The injury follows a stress fracture continuum: from bone oedema and early cortical stress reaction, through incomplete fracture, to complete bilateral pars defect.
- If bilateral pars fractures occur, the structural connection between the posterior elements and the vertebral body is severed, which — under continued loading — can allow the vertebral body to translate anteriorly, producing spondylolisthesis.
- The L5/S1 level is most commonly affected because L5 is the most lordotic segment and bears the greatest extension moment.
- In young female athletes, hypoestrogenism from RED-S impairs osteoblastic bone repair, dramatically increasing susceptibility to and slowing recovery from stress fractures at this and other sites.
- Sharp pain at L5 on extension: Hyperextension closes the posterior elements, directly compressing and shearing the fractured pars — reproducing sharp, localised pain at the stress fracture site.

■ DIFFERENTIAL DIAGNOSES

1. Normal lower limb reflexes argue against L5/S1 radiculopathy

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

■ Key Observation Findings (State These Before Tests)

Posture: Observe for hyperlordotic lumbar posture — common in gymnasts and predisposes to pars stress; note anterior pelvic tilt Active ROM: Lumbar extension painful and restricted. Flexion likely full and pain-free. Lateral flexion may reproduce pain ipsilaterally. Gait:

■ TREATMENT PLAN

- Keep knee fully extended; dorsiflex ankle passively
- Slowly raise leg from plinth, noting angle of onset of symptoms
- Positive: reproduction of radicular pain (not LBP alone) below knee at <70°
- Repeat bilaterally; compare onset angle and symptom quality
- Expected Finding in This Case:
- ■ Outcome Measures
- Numeric Pain Rating Scale (NPRS 0–10) — track pain at rest and on extension; quick, sensitive to change.
- Oswestry Disability Index (ODI)
- — measures functional disability in LBP; percentage score enables return-to-sport readiness assessment.
- Return-to-Sport Criteria Checklist
- (extension ROM normalisation, Stork test pain <1/10, psychological readiness via adapted ACL-RSI). Also monitor
- for RED-S risk at each review session.